Lesson 1 Examples of Lie algebras

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Let F denote the ground field.

Def linear algebra

Let A be a vector space over F. We call A a linear algebra if there is a binary bilinear operation $A \times A \to A$ on A, denoted by $(a,b) \mapsto ab$.

Example $\mathfrak{sl}(n)$

$\begin{cases} \text{simple} & \text{if } \operatorname{char} F = 0 \text{ or } \operatorname{char} F \nmid n \\ \text{not simple} & \text{if } \operatorname{char} F \mid n. \quad (\langle I_n \rangle \text{ is a 1-dim ideal}) \end{cases}$

Examples of Lie algebras

① from associative algebras $A \Rightarrow A^{(-)}: [a,b] = ab - ba$

Wedderburn thm in associative algebras

Suppose F is algebraically closed.

A is simple if (1) $\nexists\, I \triangleleft A$, $I \neq (0), A$. (2) $A^2 \neq (0)$

to rule out $A = \langle x \rangle$ with $x^2 = 0$ ($\Rightarrow \dim A = 1$)

(1) A simple finite dimensional associative F-algebra is isomorphic to $M_n(F)$.

(2) A fin. dim. $N \triangleleft A$; N is nilpotent, $A/N \simeq M_{n_1}(F) \oplus \cdots \oplus M_{n_k}(F)$

② from involution

Def involution

Let A be any linear algebra. A linear transformation $*: A \to A$ is called an involution if $\forall a, b \in A$

(1) $(a^*)^* = a$

(2) $(ab)^* = b^* a^*$

Examples

(1) transpose $(a_{ij}) \to (a_{ji})$

(2) quaternion

$\mathbb{H} = \{ a + bi + cj + dk \mid a,b,c,d \in \mathbb{R} \}$ with $\begin{cases} i^2 = j^2 = k^2 = -1 \\ ij = k,\ jk = i,\ ki = j \end{cases}$

$\{ij=k,\ jk=i,\ ki=j$ $\quad k \leftarrow j$
$(ji=-k,\ kj=-i,\ ik=-j)$

involution: $a+bi+cj+dk \mapsto a-bi-cj-dk$
(note that if $x=a+bi+cj+dk \neq 0$, then $x^{-1} = \frac{1}{a^2+b^2+c^2+d^2}(a-bi-cj-dk)$)
On the other hand, $x=a+bi+cj+dk$ can be represented by $\begin{pmatrix} a+bi & c+di \\ -c+di & a-bi \end{pmatrix}$
(complex 2x2 matrix)

Now the above involution has the form $\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \mapsto \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix}$

(3) nondegenerate forms. ($O(n)$, $sp(n)$. $sl(n)$).

☆ $K = \{a \in A \mid a^* = -a\} \subseteq A^{(-)}$
$([a,b]^* = b^*a^* - a^*b^* = ba - ab = -[a,b])$

③ from derivation of any linear algebra
Note that the commutator of two derivations is again a derivation
$\Rightarrow \operatorname{Der} A \subseteq \operatorname{End}_V(A)^{(-)}$

Example $W_n = \operatorname{Der} F[x_1, \cdots, x_n]$
$\operatorname{Der} F[x_1,\cdots,x_n] = \{ \sum_{i=1}^n f_i(x_1,\cdots,x_n)\frac{\partial}{\partial x_i} \mid f_i \in F[x_1,\cdots,x_n]\}$
$\Rightarrow W_n$ is infinite dimensional (W_n is in fact simple)
Consider the divergence operator:
for any $D = \sum_{i=1}^n f_i \frac{\partial}{\partial x_i} \in W_n$, the divergence is defined as $\operatorname{div}(D) = \sum_{i=1}^n \frac{\partial f_i}{\partial x_i}$
The subset $\{D \in W_n \mid \operatorname{div}(D) = 0\}$ is a simple subalgebra of W_n.
(Note that if $\operatorname{div} D_1 = \operatorname{div} D_2 = 0$, $\operatorname{div}[D_1, D_2] = 0$)

Example $ad(L)$
(1) let A be an associative algebra. Fix $a \in A$, $x \mapsto [a,x]$ is a derivation
(2) More general, let L be a Lie algebra. Then $ad(L) \lhd \operatorname{Der}(L)$
(Note that $\forall d \in \operatorname{Der}(L)$, $[d, ad(x)] = ad(d(x))$) called inner derivation

Rmk der is natural in the sense that for any $d \in$ der, if $\exp d := Id + d + \frac{d^2}{2!} + \cdots$ makes sense (e.g. d is nilpotent), then $\exp d$ is an automorphism with inverse $\exp(-d)$ (case by case, not a general thm)

$\frac{d^2}{2!} + \cdots$ makes sense (e.g. d is nilpotent), then $\exp d$ is an automorphism with inverse $\exp(-d)$. (prove case by case; not a general thm)

e.g.

(1) A finite dimensional ($d \in \mathrm{End}\, A$ can be viewed as a matrix)

(2) $A = F[[t_1, t_2]]$ $\quad d = t_1 t_2 \frac{\partial}{\partial t_1}$ $\Rightarrow \exp d\,(t_1^m t_2^n) = t_1^m t_2^n e^{m t_2}$

④ brackets

Def Poisson bracket

let A be an associative commutative algebra. A bracket $[\ ]: A \times A \to A$ is called a Poisson bracket if

Leibniz's law

① $(A, [\,,\,])$ is a Lie algebra $\quad$ i.e. the bracket is also a derivation

② $[a^2, b] = 2a[a,b]$ (In general, we require $[ac, b] = a[c,b] + [a,b]c$)

$\Leftrightarrow$ we apply $[(a+c)^2, b]$

Example

(1) $B = F(p_1, \cdots, p_n; q_1, \cdots, q_n)$

$\forall f, g \in B$, define

$$[f,g] = \sum_{i=1}^{n} \left(\frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} - \frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} \right) \quad \text{Hamilton}$$

(2) Lie but not Poisson

$B = F[t]$ $\quad \forall f, g \in B$, define $[f,g] = f'g - fg'$

⑤ Lie groups

Example solvable but not nilpotent

$$L = \begin{pmatrix} * & * \\ 0 & 0 \end{pmatrix} \subseteq gl(2)$$

basis: $\{e_{11}, e_{12}\}$ with $[e_{11}, e_{12}] = e_{12}$.